
AUDIO RECORDING LIST

OPEN STANCE

Every voice line, sound effect, ambience and music cue the game needs, by category, as a production checklist.

VERSION	UPDATED	TOTAL ITEMS	M1	M2	V1.0	FUTURE
0.4 draft	2026-10-04	169	5	31	120	13

Companion to Docs/GDD_Open_Stance.docx (v0.10), which is the source of truth; this is the audio asset list GDD 1.10 and 18.2 describe. Phases match the GDD and the motion capture shot list: v1.0 is the game the capstone delivers, with M1 and M2 the milestones that need their sounds first, and Future is everything after v1.0.

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HOW TO USE THIS LIST

CHECKBOXES

COLUMN	TICK IT WHEN
Sourced	A usable file is downloaded (licence logged), generated or recorded, and backed up.
Edited	The best files are chosen, trimmed, cleaned, layered where needed and levelled.
In engine	Imported into Unreal, named correctly, and playing from its trigger in game.

PHASES

M1 Milestone 1 (due 9 November 2026): the first score. The protector hit, kick swishes and footsteps behind the Skill Stick proof of concept (GDD 1.9, 20.1).

M2 Milestone 2 (due 7 December 2026): a full local match. Referee commands, every scoring impact, blocks and clinch, the scoreboard sounds and the competition arena crowd.

v1.0 The rest of the game the capstone delivers: fighter voices, knockdowns, every Career venue, training, Career, menus and music.

FUTURE After v1.0: commentary, video review, Referee Mode, accessibility cues, team events, para taekwondo, cadet and junior fighters, online and the world games.

HOW IT IS MADE

CODE	MEANING	ITEMS
VO	YOU RECORD. The referee commands at home: your voice for the male referees, a volunteer's for the female referees.	13
LIBRARY	DOWNLOAD. CC0 or royalty-free impacts, whooshes, footwork, gear, crowds, reactions, fighter and coach voices, room tones.	108
SYNTH	GENERATE. Synthesized by script (buzzers, UI, stingers, whooshes, hum), so there is no licence to track.	36
MUSIC	Composed or licensed (GDD 18.1); not recorded unless you compose it yourself.	12

VOICE RECORDING RULES (REFEREE COMMANDS)

- 48 kHz, 24-bit WAV, mono.
- Slate every take: say the item ID and take number before the take, for example "REF-04, take 2".
- Record 30 s of room tone at the start of every session, with nothing moving.
- At home with the AT2020USB+: a closet of clothes or blankets around you, fans and fridge off, 15–20 cm from the mic with a pop filter. Step back to arm's length, or lower the input level, for the loud reads so the mic does not overload.
- Three takes of every line per voice, at the same distance every time.
- Korean: check every read with a Korean-speaking referee or instructor before the session.
- Female referee voices: written release from your volunteer, and a guardian's consent if she is under 18.

- Keep every line E10+ (GDD 1.5): clear commands, no swearing, nothing that mocks a fighter.

SOURCING RULES (EVERYTHING ELSE)

- Anything the game plays more than once a round (impacts, footsteps, whooshes, kihaps) needs at least 5 variants, so the game can pick at random without repeats.
- Match loudness across the impact tiers (a 6 clearly bigger than a 2) and across the variants of one item.
- Log every downloaded file in the licence log (see Sources and Licences).
- Crowd beds must not contain clear referee commands or announcements, which would fight the in-game referee.

NAMING

Use `OS_<ID>_<Short name>_<Variant>_T<take>.wav` for raw files, for example `OS_REF-04_ShiJak_Kelvin_T02.wav` or `OS_IMP-05_SpinHead_V03.wav` (V is a variant number for downloaded and generated files, T a take for recordings). Unreal assets drop the take number once a take is selected and use the GDD 16.5 prefixes in Content/Audio: `SW_` for sound waves and `MS_` for MetaSounds.

SCHEDULE

Backwards from Milestone 1 (9 November) and Milestone 2 (7 December 2026). The GDD has no dates for Milestones 3 to 5 yet (GDD 20), so the work is front-loaded. You record only the referee commands; everything else is downloaded or generated, so no tournament or dojang recording is needed.

WHEN	WHERE	WHAT	ITEMS
Oct 5–16	Home	Set up the voice corner and test your mic. Start the licence log. Download the M1 and M2 library items and layer the six impact tiers. Source or generate the buzzers.	IMP-01 to 06, MOV-01 to 05, FTW-01 to 07, DEF-01 to 03, CRD-01, MAT-02/03
Sat Oct 17	Tournament (optional)	Not for recording. Meet a Korean-speaking referee or instructor who can check your pronunciation, and ask about a female volunteer for the female referee voices.	
Oct 19–Nov 8	Home	Finish the M2 library and generated items. Name, log and import them.	
Nov 9 week	—	Milestone 1 is due: no audio work.	
Nov 10–22	Home	Voice session 1: referee commands in your voice, 1–2 male referees, enough for the M2 demo. Generate the score sounds and UI.	REF-01 to 08, MAT-01/04/05/06, UI
Nov 23–Dec 6	Home	Edit, import and mix. Lock the M2 audio on 1 December and retake anything that fails.	
December	Home, volunteer	Voice session 2: the remaining male referee voices, the knockdown count and the extra calls. A one-hour session with your female volunteer, with a signed release. Download fighter and coach voice sets.	REF-09 to 13, FVO, COA
Jan 4–29	Home	Library and generated items for v1.0: remaining impacts, footwork, falls, gear, training, Career stingers and the club and school crowds.	MOV, FTW, GEA, TRG, CAR, CRD-02 to 05
Feb 1–19	Home	Build the national, continental and world-final beds by layering library crowds under a larger reverb. Source or compose the music. Audit the licence log.	CRD-06 to 08, MUS
Late Feb–spring	Home	Final mix and loudness. After beta, retake only the voice lines that fail.	

People: ask a Korean-speaking referee or instructor to check every Korean read, bring a short release form for the female volunteer (and a guardian's consent if she is under 18), and credit her in the credits if she agrees.

EQUIPMENT

Only the referee voices are recorded, so the kit is small. Prices are approximate US dollars; check them before buying.

ITEM	WHY	PRICE
Audio-Technica AT2020USB+	Referee voices at home. A USB condenser: plug it straight into your laptop, with no interface or phantom power needed. It has headphone monitoring on the mic. Set the input level in your recording software or the system sound settings, and lower it for the loud reads.	owned
Pop filter	Stops plosives on voice.	\$10–15
Closed-back headphones	Hear hum or room noise while recording; any pair works.	owned or \$30–60
iPhone (optional)	Not needed for voice. If you ever want to plug the mic into the phone, you would need a USB adapter, so check the mic's cable and what your recording app supports.	owned
Blankets or a closet of clothes	A free vocal booth: absorbs room echo.	free
Reaper or Audacity	Recording, editing, layering and loudness; Audacity is free and Reaper has a low-cost licence.	free to ~\$60

Roughly \$0–55 USD (a pop filter and, if needed, headphones). Skip the audio interface, a field recorder, a dynamic mic and a shotgun mic: nothing else here is recorded.

SOURCES AND LICENCES

Everything except the referee voices is downloaded or generated. Keep it clean for a possible commercial release (GDD 1.13).

- Download (LIBRARY items): impacts, whooshes, footwork, gear, crowds, reactions, fighter and coach voices, room tones and impulse responses. Freesound lets you filter by licence; the yearly SonniSS GDC bundles are royalty-free, but read their terms first. Paid martial-arts or fighting-game packs are fine if the licence covers games.
- CC0 is safest. CC-BY needs a credit in your credits screen. Avoid non-commercial (NC) licences, and never rip sounds from other games, films or videos.
- Voice libraries (kihap, grunts, breathing, shouts): check that the licence covers character voices in a game, and keep every one E10+ (GDD 1.5).
- Generate (SYNTH items): buzzers, UI, stingers, whooshes and hum are synthesized by script, so you own them outright. No generator script exists yet.
- AI sound-effect services each have their own terms on commercial use and ownership of output. Read them before relying on one, and log the service, prompt and date.
Do not use text-to-speech for the referee.
- Log every downloaded file below and keep a copy of its licence text.

ITEM ID	FILE AND SOURCE URL	LICENCE	DATE	CREDIT NEEDED
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐

SUMMARY BY CATEGORY

#	CATEGORY	ID PREFIX	ITEMS	M1	M2	V1.0	FUTURE
1	Referee voice	REF	13	—	8	4	1
2	Protector and scoring impacts	IMP	11	1	5	5	—
3	Kicks, punches and body movement	MOV	12	2	3	7	—
4	Footwork and falls	FTW	16	2	5	9	—
5	Defense and clinch	DEF	10	—	3	7	—
6	Fighter voice	FVO	12	—	—	12	—
7	Gear and dobok handling	GEA	7	—	—	7	—
8	Match flow and scoreboard	MAT	11	—	6	5	—
9	Crowd and venues	CRD	17	—	1	16	—
10	Coach and corner	COA	7	—	—	7	—
11	Training dojang	TRG	9	—	—	9	—
12	Career and ceremony	CAR	9	—	—	9	—
13	Menus and interface	UI	11	—	—	11	—
14	Music cues	MUS	12	—	—	12	—
15	Future	FUT	12	—	—	—	12
	Total		169	5	31	120	13

Items, not files: one item covers all its variants and takes. The referee rows alone come to more than 200 kept takes (six voices, 3 takes of every line).

1 REFEREE VOICE

The referee speaks only the Korean commands in GDD 8.5, with English subtitles on screen (GDD 7.6, 11.2). Six referees (Kelvin, Bo, Jorge, Omari, Walter and Vivian) are picked at random per match, so each needs its own voice. You record the male referee voices yourself; a female volunteer records the female referee voices. One performer can cover several referees by changing delivery and, lightly, pitch in editing, but a heavily processed voice still sounds like the same person, so the M2 demo needs only 1–2 voices. Keep 3 takes of each line per voice so repeated calls never sound identical. Calls are projected across an 8 m mat: step back for the loud reads.

PRONUNCIATION GUIDE

COMMAND	HANGUL	SAY IT	ENGLISH SUBTITLE	COMMAND	HANGUL	SAY IT	ENGLISH SUBTITLE
CHARYEOT	차렷	chah-ryuht	Attention	SET	셋	set	Three
KYEONGRYE	경례	kyung-nyeh	Bow	NET	넷	net	Four
JOON-BI	준비	joon-bee	Ready	DASEOT	다섯	dah-suht	Five
SHI-JAK	시작	shee-jahk	Start	YEOSEOT	여섯	yuh-suht	Six
KAL-YEO	갈려	kahl-yuh	Break	ILGOP	일곱	eel-gohp	Seven
KYE-SOK	계속	kyeh-sohk	Continue	YEODEOL	여덟	yuh-duhl	Eight
KEU-MAN	그만	kuh-mahn	Stop	CHUNG	청	chuhng	Blue side
GAM-JEOM	감점	gahm-juhm	Penalty	HONG	홍	hohng	Red side
HANA	하나	hah-nah	One	SEUNG	승	seung	Wins, after the side: Hong seung
DUL	둘	dool	Two	KYE-SHI	계시	kyeh-shee	Time-out (Future)

Korean has no strong word stress: give each syllable equal weight and let the call rise or snap at the end. Subtitles show the command and its meaning, for example KAL-YEO: Break (GDD 7.6).

LINES

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	REF-01	Charyeot (attention)	6 voices × 3 takes	V0	M2	Called from between the fighters, before the bow
☐	☐	☐	REF-02	Kyeongrye (bow)	6 voices × 3 takes	V0	M2	Start and end of every match
☐	☐	☐	REF-03	Joon-bi (ready)	6 voices × 3 takes	V0	M2	
☐	☐	☐	REF-04	Shi-jak (start)	6 voices × 3 takes	V0	M2	The round clock starts on this call; sharp and clear

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	REF-05	Kal-yeo (break)	Normal and urgent reads; 6 voices × 3 takes each	V0	M2	Normal: separates a clinch. Urgent: stops play for a penalty (GDD 11.2)
☐	☐	☐	REF-06	Kye-sok (continue)	6 voices × 3 takes	V0	M2	After a break, a penalty or a knockdown count
☐	☐	☐	REF-07	Keu-man (stop)	Normal (end of round) and urgent (stoppage); 6 voices × 3 takes each	V0	M2	
☐	☐	☐	REF-08	Gam-jeom (penalty)	6 voices × 3 takes	V0	M2	Said after the side, as in Hong, gam-jeom (GDD 8.4, 11.2)
☐	☐	☐	REF-09	Knockdown count: hana to yeodeol	Each number one to eight as its own file; 6 voices × 3 takes	V0	v1.0	Paced about 1 s apart; the count stops when the fighter is up (GDD 11.2)
☐	☐	☐	REF-10	Knockdown count, continuous	Hana to yeodeol in one take at match pace; 6 voices × 2 takes	V0	v1.0	Timing reference; slice it if it sounds more natural than single numbers
☐	☐	☐	REF-11	Chung / Hong (side call)	6 voices × 3 takes each	V0	v1.0	Named with a gam-jeom and in the winner call (GDD 8.5, 11.2)
☐	☐	☐	REF-12	Chung seung / Hong seung (winner)	6 voices × 3 takes each	V0	v1.0	Said while raising the winner's hand (GDD 8.5, 11.2)
☐	☐	☐	REF-13	Kye-shi (time-out)	6 voices × 3 takes	V0	FUTURE	Video review and injury stoppages (GDD 8.5, 11.2); verify against current WT rules (GDD 18.1)

2 PROTECTOR AND SCORING IMPACTS

Sourced from sound libraries (see Sources and Licences). The protector sound changes with the points scored, 1 to 6, and confirms every score (GDD 1.10, 8.3). Search martial-arts, punch and kick impacts and layer a low thud, a snappy slap and a soft cloth hit; the helmet is higher and hollower. Build the tiers so a 1 is the lightest and shortest and a 6 the biggest, with at least 5 variants each, and match loudness across tiers.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	IMP-01	Kick to the trunk protector (2 points)	Roundhouse (instep), side kick (heel), hook kick (heel); 5 variants each, medium to hard	LIBRARY	M1	The core score sound; it has to read instantly (GDD 1.9, First score)
☐	☐	☐	IMP-02	Punch to the trunk protector (1 point)	Gloved straight punch, both hands; 5 variants each	LIBRARY	M2	Duller and lighter than a kick
☐	☐	☐	IMP-03	Kick to the helmet (3 points)	Roundhouse, side, hook and axe kick (top of the helmet); 5 variants each	LIBRARY	M2	Higher and sharper than the trunk
☐	☐	☐	IMP-04	Spinning kick to the trunk (4 points)	Spin side, spin hook and tornado kick; 5 variants each, hard	LIBRARY	M2	Heavier and fuller than a 2-point kick
☐	☐	☐	IMP-05	Spinning kick to the helmet (6 points)	Spin hook, tornado and spin side kick; 5 variants each, hard	LIBRARY	M2	The biggest impact in the game; layered with the +6 score sound (MAT-01)
☐	☐	☐	IMP-06	Light contact below the impact threshold	Kicks that touch the protector or helmet without scoring; 5 variants	LIBRARY	M2	Must sound clearly weaker than a score so players know it did not count (GDD 8.3)
☐	☐	☐	IMP-07	Glancing kick	Foot sliding off the protector edge and the helmet; 5 variants each	LIBRARY	v1.0	
☐	☐	☐	IMP-08	Jump kick impacts	Jump side, roundhouse, hook, axe and spin hook kick; body and head	LIBRARY	v1.0	Same points as the grounded kick (GDD 9.2); use only if IMP-01 to IMP-05 do not fit
☐	☐	☐	IMP-09	Kick below the belt (foul)	Kick landing on the thigh or hip; 5 variants	LIBRARY	v1.0	A gam-jeom (GDD 8.4); no protector sound
☐	☐	☐	IMP-10	Punch to the helmet (foul)	Glove to helmet; 5 variants	LIBRARY	v1.0	Unscorable and a gam-jeom (GDD 8.4)
☐	☐	☐	IMP-11	Reduced-power impacts	IMP-01 to IMP-05 at about 80 % power; 3 variants each	LIBRARY	v1.0	Below 25 stamina, impact is 20 % lower (GDD 8.9). Try processing first; use only if that sounds fake

3 KICKS, PUNCHES AND BODY MOVEMENT

Whooshes and cloth snaps for each technique, from libraries (search martial arts whoosh, cloth snap, sleeve flap). Match the length to the animation flow: bounce, chamber, technique, rechamber, land (GDD 8.6); spinning kicks need a longer, rising whoosh.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	MOV-01	Roundhouse kick swish	Lead and back leg; 5 variants each	LIBRARY	M1	Includes the trouser snap at full extension
☐	☐	☐	MOV-02	Side kick / cut kick swish	Lead and back leg; 5 variants each	LIBRARY	M1	
☐	☐	☐	MOV-03	Hook kick swish	Lead and back leg; 5 variants each	LIBRARY	M2	
☐	☐	☐	MOV-04	Spinning kick swish	Spin side kick and spin hook kick; 5 variants each	LIBRARY	M2	Longer, rising swish with the turn
☐	☐	☐	MOV-05	Punch swish	Straight punch, both hands; 5 variants each	LIBRARY	M2	Short sleeve snap
☐	☐	☐	MOV-06	Axe kick swish	Leg rising and heel dropping; 5 variants	LIBRARY	v1.0	
☐	☐	☐	MOV-07	Tornado kick swish	5 variants	LIBRARY	v1.0	The longest swish, for the hardest technique (GDD 9.2)
☐	☐	☐	MOV-08	Jump takeoff	One-foot and two-foot push-off; 5 variants each	LIBRARY	v1.0	Jump kicks share one launch-and-land layer (GDD 16.4)
☐	☐	☐	MOV-09	Fake attack	Chamber starts and pulls back with no extension; 5 variants	LIBRARY	v1.0	Startup only (GDD 8.6)
☐	☐	☐	MOV-10	Chamber and rechamber	Knee lift and retraction, cloth only; lead and back leg	LIBRARY	v1.0	
☐	☐	☐	MOV-11	Guard change	High, cross, mixed and low; sleeve and glove rustle	LIBRARY	v1.0	D-pad guard switch (GDD 7.3)
☐	☐	☐	MOV-12	Dobok and gear rustle in guard	Bouncing in guard; 30 s loops, relaxed and working	LIBRARY	v1.0	Sits under the footwork bounce

4 FOOTWORK AND FALLS

Feet squeak and slap on the mat (GDD 1.10). Search sneaker squeaks on gym floors, mat and foam footsteps, and body falls on mats. Falls and landings need a heavier body-and-cloth hit.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	FTW-01	Bounce in place	2 bounces per second; 30 s loops, fresh and tired	LIBRARY	M1	Values from GDD 6.4
☐	☐	☐	FTW-02	Step	Forward, back and side, 40 cm; 10 variants each	LIBRARY	M1	
☐	☐	☐	FTW-03	Advance and retreat (held)	Continuous loops, forward and back	LIBRARY	M2	1.2 m/s (GDD 6.4)
☐	☐	☐	FTW-04	Dash	Forward and back; 5 variants each	LIBRARY	M2	Quick push, skid and plant (GDD 9.3)
☐	☐	☐	FTW-05	Pivot squeak	Supporting-foot pivots for kicks and spins; 10 variants	LIBRARY	M2	The squeak under a spinning kick
☐	☐	☐	FTW-06	Switch sides	Quick stance-change shuffle; 5 variants	LIBRARY	M2	GDD 8.7
☐	☐	☐	FTW-07	Kick landing	Forward and back landing after a kick; 5 variants each	LIBRARY	M2	
☐	☐	☐	FTW-08	Side step and circle	Left and right; 5 variants each	LIBRARY	v1.0	
☐	☐	☐	FTW-09	Jump landing	One-foot and two-foot; light and heavy	LIBRARY	v1.0	
☐	☐	☐	FTW-10	Stagger	2–3 uneven steps after a heavy hit; 5 variants	LIBRARY	v1.0	Heavy hit reaction (GDD 8.10)
☐	☐	☐	FTW-11	Knockdown fall	Onto the back, side and front; body, gear and dobok hitting the mat	LIBRARY	v1.0	
☐	☐	☐	FTW-12	Failed landing	Slip or stumble into a fall after a spinning or jump kick; 3 variants	LIBRARY	v1.0	The Falling penalty path (GDD 8.10)
☐	☐	☐	FTW-13	Get-up	From the back and side, back to guard; 3 variants each	LIBRARY	v1.0	
☐	☐	☐	FTW-14	Walk on and off the mat	To the start mark; winner and loser walk-offs	LIBRARY	v1.0	Match intro and results
☐	☐	☐	FTW-15	Referee footsteps	White sneakers on the mat: walk, quick step in, step back; loops	LIBRARY	v1.0	Different from the fighters' feet (GDD 11.2)
☐	☐	☐	FTW-16	Wood-floor footsteps	Walk, bounce and step on a wood floor	LIBRARY	v1.0	Training dojang (GDD 6.2); only if fighters train off the mat there

5 DEFENSE AND CLINCH

Contact that is not a score: blocks, checks, clinches and the touches of the sport's etiquette. Layer glove, forearm and cloth impacts from libraries.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	DEF-01	Arm block of a body kick	Forearm and glove absorbing a roundhouse; 5 variants	LIBRARY	M2	Active Defense, LT alone (GDD 8.8)
☐	☐	☐	DEF-02	High guard block	Head kick stopped on the forearms; 5 variants	LIBRARY	M2	LT + RS Up (GDD 8.8)
☐	☐	☐	DEF-03	Clinch contact	Chest protectors pressing, dobok rustle, gloves on shoulders; 10 s loops	LIBRARY	M2	Chest-to-chest contact starts a clinch (GDD 6.4)
☐	☐	☐	DEF-04	Leg check	Shin and knee stopping a body kick; 5 variants	LIBRARY	v1.0	LT + RS Down (GDD 8.8)
☐	☐	☐	DEF-05	Sway or step off the line	Fast footwork and cloth; 5 variants	LIBRARY	v1.0	LT + RS Left/Right (GDD 8.8)
☐	☐	☐	DEF-06	Punch blocked	Glove on glove and glove on forearm; 5 variants	LIBRARY	v1.0	
☐	☐	☐	DEF-07	Kick clash	Two kicks meeting, shin to shin and foot to foot; 5 variants	LIBRARY	v1.0	Both fighters kick at once
☐	☐	☐	DEF-08	Grab and push	Dobok grabbed and shoved; 5 variants	LIBRARY	v1.0	A gam-jeom (GDD 8.4)
☐	☐	☐	DEF-09	Separation	Fighters stepping apart after kal-yeo	LIBRARY	v1.0	
☐	☐	☐	DEF-10	Glove touch	Gloves tapped together; 3 variants	LIBRARY	v1.0	Start and end of the match

6 FIGHTER VOICE

Fighters kihap (shout) as they kick, grunt when hit and breathe harder as their stamina drops below 25 (GDD 1.10, 8.9); they play on the fighter and coach voice slider (GDD 7.8). Download from voice-effort libraries (kihap, martial-arts shouts, grunts, breathing), with at least two men's and two women's sets so the two fighters in a match sound different. Check each licence covers games and character voices. Keep everything E10+ (GDD 1.5): efforts and shouts, no pain screams, nothing that mocks a fighter.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	FV0-01	Kihap, short	Sharp one-syllable shout; 10 variants per voice set	LIBRARY	v1.0	On kick extension or right after a score
☐	☐	☐	FV0-02	Kihap, long	Held shout after a big score; 5 variants per voice set	LIBRARY	v1.0	Spinning and head kicks
☐	☐	☐	FV0-03	Kick exhale	Short sharp breath, no voice; 10 variants per voice set	LIBRARY	v1.0	Under every kick; quieter than a kihap
☐	☐	☐	FV0-04	Punch exhale	10 variants per voice set	LIBRARY	v1.0	
☐	☐	☐	FV0-05	Hit grunt, light	Air pushed out by a body kick; 5 variants per voice set	LIBRARY	v1.0	Minor and normal reactions (GDD 8.10)
☐	☐	☐	FV0-06	Hit grunt, heavy	5 variants per voice set	LIBRARY	v1.0	Heavy and knockdown reactions (GDD 8.10)
☐	☐	☐	FV0-07	Fall and get-up efforts	Landing breath and the push to stand; 3 variants per voice set	LIBRARY	v1.0	
☐	☐	☐	FV0-08	Breathing in guard	Fresh, working and tired; 30 s loops per voice set	LIBRARY	v1.0	The tired loop plays below 25 stamina (GDD 8.9)
☐	☐	☐	FV0-09	Breathing in the corner	Recovering, 60 s loop; drinking water	LIBRARY	v1.0	1:00 round break (GDD 8.2)
☐	☐	☐	FV0-10	Score reaction	Quiet "yes" or sharp exhale; 5 variants per voice set	LIBRARY	v1.0	Controlled, never at the opponent
☐	☐	☐	FV0-11	Frustration after conceding	Exhale or short groan, no words; 5 variants per voice set	LIBRARY	v1.0	
☐	☐	☐	FV0-12	Victory shout	Big win in a final; 3 variants per voice set	LIBRARY	v1.0	Career finals and the podium

7 GEAR AND DOBOK HANDLING

Handling sounds for the gear-up sequence, the round break and the menus, where the fighter appears in dobok and belt (GDD 4, 13), and for the creator's gear on and off preview (GDD 7.5, 10). Search Velcro, strap, clip, cloth and bag sounds in libraries.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	GEA-01	Chest protector on and fastened	Straps and Velcro; 3 variants	LIBRARY	v1.0	Gear-up sequence
☐	☐	☐	GEA-02	Helmet on and off	3 variants each	LIBRARY	v1.0	Off after the match
☐	☐	☐	GEA-03	Gloves and foot guards on	Velcro and sock pull; 3 variants each	LIBRARY	v1.0	
☐	☐	☐	GEA-04	Gear adjusted	Helmet, protector straps and gloves; short	LIBRARY	v1.0	Between rounds
☐	☐	☐	GEA-05	Belt tied and tugged	3 variants	LIBRARY	v1.0	Menus and the creator
☐	☐	☐	GEA-06	Dobok handled	Jacket put on, straightened and smoothed	LIBRARY	v1.0	Menus in dobok
☐	☐	☐	GEA-07	Gear bag	Zip, drop on a bench, rummage	LIBRARY	v1.0	Locker-room menus (GDD 13)

8 MATCH FLOW AND SCOREBOARD

The electronic side of the match: the score confirmation that plays with the score pop-up (GDD 2.2, 7.6) and the scoreboard's buzzers. Synthesized by script, so they need no licence.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	MAT-01	Score confirmation, by points	+1, +2, +3, +4 and +6, plus a separate +1 for an opponent's gam-jeom	SYNTH	M2	Rises with the points, so a 6 sounds biggest (GDD 2.5, 8.3)
☐	☐	☐	MAT-02	Round start	Buzzer or tone on shi-jak	SYNTH	M2	Real PSS tone optional if you hear one at an event
☐	☐	☐	MAT-03	Round end	Buzzer when the clock reaches 0:00	SYNTH	M2	Plays before keu-man
☐	☐	☐	MAT-04	Gam-jeom marker	Penalty count added under a score	SYNTH	M2	HUD penalties (GDD 7.6)
☐	☐	☐	MAT-05	Round won	Normal win and a bigger version for the 12-point gap	SYNTH	M2	GDD 8.2
☐	☐	☐	MAT-06	Match won	Stinger as the referee raises the winner's hand	SYNTH	M2	
☐	☐	☐	MAT-07	Final 10 seconds	Subtle pulse for the last 10 s of a round	SYNTH	v1.0	The only clock sound; rounds have no music (GDD 1.10)
☐	☐	☐	MAT-08	Round break warning	10 s before the next round	SYNTH	v1.0	1:00 break (GDD 8.2)
☐	☐	☐	MAT-09	Spinning-kick camera cut	Whoosh into the 1.5 s cut	SYNTH	v1.0	GDD 13
☐	☐	☐	MAT-10	Bullet Time	Slow-motion orbit: time-stretched swish and low boom	SYNTH	v1.0	Easter egg (GDD 6.5)
☐	☐	☐	MAT-11	Arcade score set	8-bit versions of MAT-01 to MAT-06	SYNTH	v1.0	Konami Code easter egg (GDD 6.5)

9 CROWD AND VENUES

Crowd size and reverb grow with the venue, from a quiet club night to a full arena (GDD 1.10, 6.2). Download the beds, reactions and impulse responses from CC0 or royalty-free libraries (see Sources and Licences). The national, continental and world-final beds are built by layering crowds under a larger reverb. Each venue needs a 2-minute stereo bed loop; reactions come in three sizes: small (club, competition arena), medium (school gym, provincial hall) and large (national arena and up). Beds must not contain clear referee commands or announcements, which would fight the in-game referee. Crowds cheer, gasp and applaud; they never boo or mock a fighter (GDD 1.10, 6.5).

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	CRD-01	Competition arena bed	Small crowd murmur; 2 min loop	LIBRARY	M2	Default venue for Sparring and Tournament
☐	☐	☐	CRD-02	Training dojang room tone	Daylight hall, ventilation, distant traffic; 2 min loop	LIBRARY	v1.0	No crowd
☐	☐	☐	CRD-03	Club dojang bed	10–20 club mates on benches: chatter and claps; 2 min loop	LIBRARY	v1.0	Career tier 1
☐	☐	☐	CRD-04	School gym bed	A few hundred on folding bleachers, gym echo; 2 min loop	LIBRARY	v1.0	Career tier 2
☐	☐	☐	CRD-05	Provincial hall bed	Several mats running at once: distant referee calls, buzzers, coaches, PA; busy warm-up area	LIBRARY	v1.0	Career tier 3; announcements unintelligible
☐	☐	☐	CRD-06	National arena bed	Arena bowl with a full lower tier; 2 min loop	LIBRARY	v1.0	Career tier 4; layer and add reverb if no real arena
☐	☐	☐	CRD-07	Continental arena bed	Larger and louder; flags and wordless chants; 2 min loop	LIBRARY	v1.0	Career tier 5; build by layering
☐	☐	☐	CRD-08	World-final stage bed	Full crowd in a darkened arena, anticipation; 2 min loop	LIBRARY	v1.0	Career tier 6; build by layering
☐	☐	☐	CRD-09	Score cheer	Small, medium and big cheers; per crowd size	LIBRARY	v1.0	Bigger for higher points
☐	☐	☐	CRD-10	Gasp	Near miss, big spinning attempt, knockdown; per crowd size	LIBRARY	v1.0	
☐	☐	☐	CRD-11	Groan	Missed chance, penalty; per crowd size	LIBRARY	v1.0	Never aimed at a fighter
☐	☐	☐	CRD-12	Applause	Bow, match start, winner declared; per crowd size	LIBRARY	v1.0	
☐	☐	☐	CRD-13	Rhythmic clapping and stomping	Building in a close final round; medium and large	LIBRARY	v1.0	The comeback (GDD 1.9)
☐	☐	☐	CRD-14	Entrance walk swell	Crowd rising as the fighters walk out	LIBRARY	v1.0	World-final stage (GDD 6.2)

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	CRD-15	Close supporters' shouts	Generic English: "Go!", "Kick!", "Yes!", "Come on!"; 4 voices	LIBRARY	v1.0	Club dojang and school gym, where the crowd is close
☐	☐	☐	CRD-16	Officials' tables	Hand-held scoring device clicks, recorder keyboard, quiet talk; 1 min loop	LIBRARY	v1.0	Judges' and recorder tables (GDD 6.3)
☐	☐	☐	CRD-17	Venue impulse responses	Impulse responses for a dojang, school gym, sports hall and arena	LIBRARY	v1.0	Convolution reverb, so reverb grows with the venue (GDD 1.10)

10 COACH AND CORNER

A coach sits in each corner chair (GDD 6.3) and shouts short English cues during play; advice between rounds stays as on-screen text (GDD 1.10, 4.3). Use library shout and crowd-voice clips, one male and one female if the library allows; claps, bottles and chairs are library foley.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	COA-01	Coach shouts during play	Generic English: "Go!", "Again!", "Distance!", "Watch the clock!", "Good!"; 2 voices × 3 variants	LIBRARY	v1.0	GDD 1.10, 4.3
☐	☐	☐	COA-02	Coach clapping	Single claps and bursts	LIBRARY	v1.0	
☐	☐	☐	COA-03	Coach celebration	"Yes!" with claps; 2 voices	LIBRARY	v1.0	
☐	☐	☐	COA-04	Coach frustration	Exhale or short groan, no words; 2 voices	LIBRARY	v1.0	Respectful
☐	☐	☐	COA-05	Between-round murmur	Low, unintelligible advice; 30 s per voice	LIBRARY	v1.0	Mumbled, so no script is needed
☐	☐	☐	COA-06	Water bottle and towel	Squeeze, cap, swig, towel wipe	LIBRARY	v1.0	Round break
☐	☐	☐	COA-07	Corner chair	Sit, stand, scrape on the floor	LIBRARY	v1.0	

11 TRAINING DOJANG

Training lessons, drills and free practice in the dojang (GDD 3.2, 6.2): kicking paddles, reaction targets, heavy bags and a training dummy (GDD 11.3). Paddle pops, bag hits and jump-rope loops are common library sounds.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	TRG-01	Kicking paddle hit	Single and double paddle; body and head height; 10 variants each	LIBRARY	v1.0	The classic paddle pop
☐	☐	☐	TRG-02	Heavy bag hits	Roundhouse, side and spin side kick, punch; chain swing	LIBRARY	v1.0	
☐	☐	☐	TRG-03	Wall pad hits	Side kick and push kick into a wall pad	LIBRARY	v1.0	Only if the wall pads take hits
☐	☐	☐	TRG-04	Training dummy	Base rocking and settling after a hit	LIBRARY	v1.0	If the dummy wears a protector, its hits reuse IMP-01 to IMP-05
☐	☐	☐	TRG-05	Reaction target	Light-up cue and hit-confirm tone	SYNTH	v1.0	Targets light up (GDD 11.3)
☐	☐	☐	TRG-06	Drill grade	One stinger per grade	SYNTH	v1.0	Drills with grades (GDD 3.2)
☐	☐	☐	TRG-07	Lesson step complete	Short confirm when a tutorial prompt is met	SYNTH	v1.0	GDD 7.7
☐	☐	☐	TRG-08	Jump rope	Rope whip and foot taps; 30 s loops at two speeds	LIBRARY	v1.0	Training shots in the mocap list
☐	☐	☐	TRG-09	Wax On, Wax Off	Rag wiping a mirror in slow circles; squeak loop	LIBRARY	v1.0	Easter egg (GDD 6.5)

12 CAREER AND CEREMONY

Career's calendar, grading, rank and podium moments (GDD 4, 5). The stingers are synthesized; the medal and belt rows are library Foley.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	CAR-01	Grading: belt presented	Belt handed over and tied; polite club applause	LIBRARY	v1.0	Rank promotion (GDD 5.3)
☐	☐	☐	CAR-02	Rank up	Stinger for passing a grading	SYNTH	v1.0	
☐	☐	☐	CAR-03	Level up	XP level stinger	SYNTH	v1.0	GDD 5.2
☐	☐	☐	CAR-04	Technique unlocked	Stinger	SYNTH	v1.0	GDD 5.5
☐	☐	☐	CAR-05	Medal placed	Medal clink and ribbon over the head; gold, silver and bronze	LIBRARY	v1.0	Podium
☐	☐	☐	CAR-06	Calendar week advance	Page turn or day tick	SYNTH	v1.0	GDD 4.1
☐	☐	☐	CAR-07	Event entry confirmed	Stinger	SYNTH	v1.0	
☐	☐	☐	CAR-08	Bracket advance	A name moving to the next round	SYNTH	v1.0	Career and local Tournament (GDD 7.5)
☐	☐	☐	CAR-09	Locker-room ambience	Bench, lockers, distant hall noise; 2 min loop	LIBRARY	v1.0	Menu setting (GDD 13)

13 MENUS AND INTERFACE

Interface sounds for every screen in GDD 7.5. Synthesized by script, so they are free of licences; keep them short and soft so they sit under the music.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	UI-01	Move focus	Up, down, left and right	SYNTH	v1.0	
☐	☐	☐	UI-02	Confirm		SYNTH	v1.0	
☐	☐	☐	UI-03	Back		SYNTH	v1.0	
☐	☐	☐	UI-04	Tab switch	LB and RB	SYNTH	v1.0	Main menu tabs (GDD 7.5)
☐	☐	☐	UI-05	Slider tick	Volume and dead-zone sliders	SYNTH	v1.0	GDD 7.8
☐	☐	☐	UI-06	Toggle	On and off	SYNTH	v1.0	
☐	☐	☐	UI-07	Not available	Locked technique or option	SYNTH	v1.0	
☐	☐	☐	UI-08	Press any button	Title screen start	SYNTH	v1.0	
☐	☐	☐	UI-09	Pause open and close		SYNTH	v1.0	
☐	☐	☐	UI-10	Fighter select lock-in	Chung side and Hong side	SYNTH	v1.0	
☐	☐	☐	UI-11	Results tally	Ticks as points by technique count up	SYNTH	v1.0	Results screen (GDD 7.5)

14 MUSIC CUES

Music plays in menus, loading screens and round breaks only; rounds have no music (GDD 1.10). Style: modern electronic with Korean percussion. Whether it is commissioned or licensed is still open (GDD 18.1), so these are cues to source, not to record, unless you compose them yourself.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	MUS-01	Title and main menu theme	2–3 min loop	MUSIC	v1.0	
☐	☐	☐	MUS-02	Setup and fighter select	Loop	MUSIC	v1.0	
☐	☐	☐	MUS-03	Loading screens	Loop	MUSIC	v1.0	
☐	☐	☐	MUS-04	Round break	1:00 cue, calm with a pulse; ends on the next shi-jak	MUSIC	v1.0	GDD 8.2
☐	☐	☐	MUS-05	Results	Win stinger, loss stinger and a results loop	MUSIC	v1.0	
☐	☐	☐	MUS-06	Training	Loop for the dojang menus	MUSIC	v1.0	
☐	☐	☐	MUS-07	Career hub and calendar	Loop	MUSIC	v1.0	
☐	☐	☐	MUS-08	Tournament bracket	Loop between matches	MUSIC	v1.0	
☐	☐	☐	MUS-09	Entrance walk	World-final stage walk-out	MUSIC	v1.0	Career tier 6 (GDD 6.2)
☐	☐	☐	MUS-10	Podium and medal ceremony	World series final	MUSIC	v1.0	No real anthems; events are fictional (GDD 4.2)
☐	☐	☐	MUS-11	Credits		MUSIC	v1.0	
☐	☐	☐	MUS-12	Korean percussion stems	Buk, janggu and kkwaenggwari hits and loops	LIBRARY	v1.0	Only if composing in-house; also good source for stingers

15 FUTURE

Audio for features after v1.0 (GDD 17). Listed now so the v1.0 sessions can leave room for them.

SOURCED	EDITED	IN ENGINE	ID	SOUND	VARIANTS NEEDED	HOW	PHASE	NOTES
☐	☐	☐	FUT-01	Play-by-play commentary	English; needs a full script first	LIBRARY	FUTURE	Commentary (GDD 13); the largest voice job
☐	☐	☐	FUT-02	Colour commentary	English; needs a full script first	LIBRARY	FUTURE	GDD 13
☐	☐	☐	FUT-03	Video review	Coach's review card, review sting, replay rewind	SYNTH	FUTURE	GDD 8.11
☐	☐	☐	FUT-04	Referee Mode rating	Warning tone below 40; removal at 20	SYNTH	FUTURE	GDD 3.5.1; the technical delegate stays off screen
☐	☐	☐	FUT-05	Accessibility cues	Incoming attack cue; stamina state cue	SYNTH	FUTURE	GDD 7.9
☐	☐	☐	FUT-06	Team Championship	Tag in and out, team bench shouts, team score	LIBRARY	FUTURE	GDD 3.1
☐	☐	☐	FUT-07	Para taekwondo foley	Footwork, falls and contact for para athletes	LIBRARY	FUTURE	GDD 3.3; developed with para practitioners
☐	☐	☐	FUT-08	Cadet and junior voice sets	Younger voices for the fighter voice rows	LIBRARY	FUTURE	GDD 4.4; performers under 18 need a guardian's consent
☐	☐	☐	FUT-09	World games stadium	Largest crowd, opening ceremony, podium	LIBRARY	FUTURE	Career tier 7 (GDD 6.2)
☐	☐	☐	FUT-10	Online interface	Match found, rematch, disconnect	SYNTH	FUTURE	GDD 12
☐	☐	☐	FUT-11	Replays and photo mode	Rewind, slow motion, shutter	SYNTH	FUTURE	GDD 3.4
☐	☐	☐	FUT-12	Hall of fame and retirement	Ceremony music and applause	MUSIC	FUTURE	GDD 4.4

VOICE SESSION LOG

One row per session. Note the location, performers, gear and any problems, so retakes can match the original sound.

DATE	LOCATION	PERFORMERS	ITEM IDS RECORDED	MIC AND ROOM	ROOM TONE	NOTES
					☐	
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